15. Implement Iris Flower Classification using Naive Bayes classifier

Aim

To implement **Iris Flower Classification using the Naive Bayes classifier**.

Software Required

- Python 3.x
- Scikit-learn
- NumPy

Procedure

1. Load the Iris dataset.
2. Divide the dataset into training and testing sets.
3. Create a Gaussian Naive Bayes classifier.
4. Train the model.
5. Predict flower classes.
6. Calculate the accuracy.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score
iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = GaussianNB()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted Classes:")
```

```
print(y_pred)

print("\nAccuracy:",

    accuracy_score(y_test, y_pred))

sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(sample)[0]

print("\nFlower:",

    iris.target_names[prediction])
```

Output:

```
Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]

Accuracy: 1.0

Flower: setosa
```

Result

Thus, Iris flower classification was successfully implemented using the **Naive Bayes classifier**.